



Department of Physical Sciences
Faculty of Applied Sciences
Rajarata University of Sri Lanka, Mihintale
B.Sc. in Applied Sciences

STA 1202 – Introduction to Probability Theory
Tutorial No: 4

1. Upon studying low bids for shipping contracts, a microcomputer manufacturing company finds that intrastate contracts have low bids that are uniformly distributed between 20 and 25, in units of thousands of dollars. Find the probability that the low bid on the next intrastate shipping contract.
 - a. is below \$22,000.
 - b. is in excess of \$24,000
2. A telephone call arrived at a switchboard at random within a one-minute interval. The switch board was fully busy for 15 seconds into this one-minute period. What is the probability that the call arrived when the switchboard was not fully busy?
3. Let Z denote a normal random variable with mean 0 and standard deviation 1.
 - a. Find $P(Z > 2)$.
 - b. Find $P(-2 \leq Z \leq 2)$.
 - c. Find $P(0 \leq Z \leq 1.73)$.
4. The achievement scores for a college entrance examination are normally distributed with mean 75 and standard deviation 10. What fraction of the scores lies between 80 and 90?
5. One method of arriving at economic forecasts is to use a consensus approach. A forecast is obtained from each of a large number of analysts; the average of these individual forecasts is the consensus forecast. Suppose that the individual 1996 January prime interest-rate forecasts of all economic analysts are approximately normally distributed with mean 7% and standard deviation 2.6%. If a single analyst is randomly selected from among this group, what is the probability that the analyst's forecast of the prime interest rate will
 - a. exceed 11%?
 - b. be less than 9%?

6. Scores on a standardized college entrance examination (CEE) are normally distributed with mean 510 and standard deviation 60. A selective university considers for admission only applicants with CEE scores over 650. Find percentage of all individuals who took the CEE who meet the university's CEE requirement for consideration for admission.
7. Achievement test scores of all high school seniors in a state have mean 60 and variance 64. A random sample of $n = 100$ students from one large high school had a mean score of 58. Calculate the probability that the sample mean is at most 58 when $n = 100$.
8. The service times for customers coming through a checkout counter in a retail store are independent random variables with mean 1.5 minutes and variance 1.0. Approximate the probability that 100 customers can be served in less than 2 hours of total service time.

(Please answer all questions and submit the answer script to the tutorial room on or before 31st of January 2024)